

PROPOSITIONAL LOGIC - SYNTAX -

The *syntax* introduces the entities used to define well-formed propositional formulas.

- $\Sigma_P = Var_propos \cup \{F, T\} \cup Connectives \cup \{(\,)\}$ is the *vocabulary*;
- $Var_propos = \{p, q, r, \dots\}$ is a finite *set of propositional variables*;
- $Connectives = \{ \neg \text{ (negation)}, \wedge \text{ (conjunction)}, \vee \text{ (disjunction)}, \rightarrow \text{ (implication)}, \leftrightarrow \text{ (equivalence)} \}$.

The *negation* is a unary connective and all the others are binary connectives.

The decreasing order of precedence of the connectives is as follows:

$$\neg, \wedge, \vee, \rightarrow, \leftrightarrow.$$

- F_P is the *set of well-formed formulas* built using the propositional variables, the connectives and the parentheses (to avoid ambiguity).

example: $(p \rightarrow \neg q) \wedge (r \vee q \leftrightarrow p) \wedge s$

Semantics of propositional logic

- Logical propositions are models of propositional assertions from natural language, which can be true or false.
- The aim of the semantics is to give a meaning (to assign a truth value) to the propositional formulas
- The semantic domain is the set of truth values: $\{ F \text{ (false)}, T \text{ (true)} \}$, satisfy the relations:

$$\neg F = T, \neg T = F.$$

- New connectives $\uparrow$ ("nand"), $\downarrow$ ("nor"), $\oplus$ ("xor") are introduced:

$$p \uparrow q := \neg(p \wedge q), \quad p \downarrow q := \neg(p \vee q), \quad p \oplus q := \neg(p \leftrightarrow q)$$

- These new connectives are used in the design of logic circuits.
- The semantics of the connectives are provided by the following truth tables:

p	q	$\neg p$	$p \wedge q$	$p \vee q$	$p \rightarrow q$	$p \leftrightarrow q$	$p \uparrow q$	$p \downarrow q$	$p \oplus q$
T	T	F	T	T	T	T	F	F	F
T	F	F	F	T	F	F	T	F	T
F	T	T	F	T	T	F	T	F	T
F	F	T	F	F	T	T	T	T	F

Truth tables

- A conjunction is true exactly when both its operands are true.

As a generalization, the conjunction $p_1 \wedge p_2 \wedge \dots \wedge p_n$ is true exactly when all its n operands are true.

- A disjunction (“*inclusive or*”) is false only when both its operands are false.

As a generalization, the disjunction $p_1 \vee p_2 \vee \dots \vee p_n$ is false only when all its n operands are false.

- The implication $p \rightarrow q$ is false only when the hypothesis p is true and the conclusion q is false (*true cannot imply false*).

- The equivalence $p \leftrightarrow q$ is true only when p and q have the same truth value.

- The connective $\oplus$ (“*exclusive or*”) is the negation of equivalence and it is true only when one operand is true and the other one is false.

Stylistic variants in English for logical connectives

$A \wedge B$	$A \vee B$	$A \rightarrow B$	$A \leftrightarrow B$
A and B Both A and B A, but B A, although B A as well as B A, B A, also B	A or B Either A or B A unless B	If A, then B If A, B A is a sufficient condition for B A is sufficient for B In case A, B Provided that A, then B B provided that A B is necessary for A A only if B B if A	A if and only if B A is equivalent to B A is necessary and sufficient for B A just in case B

Sufficiency *versus* necessity

Example:

p: n is divisible by 4.

q: n is even.

r: n is divisible by 2.

X: If n is divisible by 4 then n is even: $p \rightarrow q$

p is **sufficient** for q, but it is **not necessary**.

q is **necessary** for p, but it is **not sufficient**.

Z: n is even if and only if n is divisible by 2.

q is **necessary** and **sufficient** for r: $q \leftrightarrow r$.

Conditional rules

p: It is sunny.

q: I will go for a walk.

X: If it is sunny then I will go for a walk.

p is sufficient for q: $p \rightarrow q$

Y: Only if it is sunny then I will go for a walk.

p is necessary for q: $\neg p \rightarrow \neg q \equiv q \rightarrow p$

Z: I will go for a walk if and only if it is sunny.

p is necessary and sufficient for q: $p \leftrightarrow q$

p	q	$p \rightarrow q$	$\neg p \rightarrow \neg q$	$q \rightarrow p$	$p \leftrightarrow q$
T	T	T	T	T	T
T	F	F	T	F	F
F	T	T	F	T	F
F	F	T	T	F	T

$$q \rightarrow p \equiv \neg p \rightarrow \neg q$$

Conditional rules are just like game rules, with events that can be true:

“if” something else is true ----- sufficiency

or “only if” something else is true ----- necessity

Conditional rule: $a \rightarrow b$

A **sufficient** condition (p) **guarantees** the truth of another condition (q), but it is **not necessary** for that other condition (q) to happen: p is sufficient for q. $p \rightarrow q$

A **necessary** condition (p) **is required** for something else (q) to happen, but it **does not guarantee** that something else happens (q), it is impossible to have q without p:

p is necessary for q. $\neg p \rightarrow \neg q \equiv q \rightarrow p$

Interpretation of a propositional formula

Definition

An *interpretation* of a formula $U(p_1, p_2, \dots, p_n) \in F_P$ is a function

$$i: \{p_1, p_2, \dots, p_n\} \rightarrow \{F, T\}$$

that can be extended to $i: F_P \rightarrow \{F, T\}$ using the following relations:

$$\begin{aligned} i(\neg p) &= \neg i(p), & i(p \wedge q) &= i(p) \wedge i(q), & i(p \vee q) &= i(p) \vee i(q) \\ i(p \rightarrow q) &= i(p) \rightarrow i(q), & i(p \leftrightarrow q) &= i(p) \leftrightarrow i(q) \end{aligned}$$

- A formula $U(p_1, p_2, \dots, p_n) \in F_P$ has 2^n *interpretations*.
- Interpretations assign truth values to propositional variables and using the semantics of the connectives evaluate formulas assigning truth values to them.
- The semantics is compositional, meaning that the truth value of a formula is obtained from the truth values of its subformulas.

Semantic concepts

Let $U(p_1, p_2, \dots, p_n)$ be a propositional formula.

1. An interpretation i which evaluates the formula U as *true* is called a **model** of U .

$$i: \{p_1, \dots, p_n\} \rightarrow \{T, F\} \text{ such that } i(U) = T.$$

2. An interpretation i which evaluates the formula U as *false* is called an **anti-model** of U :

$$i: \{p_1, \dots, p_n\} \rightarrow \{T, F\} \text{ such that } i(U) = F.$$

3. A **formula** U is called **consistent (satisfiable)** if it has a model:

$$\exists i: \{p_1, \dots, p_n\} \rightarrow \{T, F\} \text{ such that } i(U) = T.$$

4. The **formula** U is called **valid (tautology)** and we use the notation: $\models U$, if U is evaluated as true in all interpretations:

$$\forall i: \{p_1, \dots, p_n\} \rightarrow \{T, F\}, i(U) = T. \text{ All interpretations of } U \text{ are } \mathbf{models} \text{ for } U.$$

5. The **formula** U is called **inconsistent** if U does not have any model,

$$U \text{ is evaluated as false in all interpretations: } \forall i: \{p_1, \dots, p_n\} \rightarrow \{T, F\}, i(U) = F.$$

6. The **formula** U is called **contingent** if U is consistent, but is not valid.

Semantic concepts (contd.)

The *logical consequence* notion is a generalization of the *tautology* notion:

Definition: The formula V is a *logical consequence* of the formula U ,

notation: $U \models V$,

if $\forall i : F_P \rightarrow \{T, F\}$ such that $i(U)=T$, we have $i(V)=T$.

Definition: The formulas U and V are *logically equivalent*, notation: $U \equiv V$,

if they have identical truth tables.

Note:

“ $\models$ ” and “ $\equiv$ ” are *meta-symbols* used to express *logical relations* between formulas.

Problems in propositional logic

- Check the ***validity*** / ***consistency*** / ***inconsistency*** property of a propositional formula;
- Find the ***models*** and ***anti-models*** of a consistent formula
- Check the ***logical equivalence*** and ***logical consequence*** relations between two propositional formulas
- Check the ***logical consequence*** relation between a ***set of premises (hypotheses)*** and a ***conclusion***.

Example 1. Build the truth tables of the formulas:

$$U(p,q,r) = (\neg p \vee q) \wedge (r \vee p), \quad V(p,q,r) = (\neg p \wedge r) \vee (q \wedge r) \vee (q \wedge p),$$

$$W(p,q,r) = (p \uparrow (\neg p \wedge q)) \vee r, \quad Z(p,q,r) = p \wedge ((\neg q \vee r) \downarrow q).$$

	p	q	r	$\neg p \vee q$	$r \vee p$	$U(p,q,r)$	$V(p,q,r)$	$W(p,q,r)$	$Z(p,q,r)$
i1	T	T	T	T	T	T	T	T	F
i2	T	T	F	T	T	T	T	T	F
i3	T	F	T	F	T	F	F	T	F
i4	T	F	F	F	T	F	F	T	F
i5	F	T	T	T	T	T	T	T	F
i6	F	T	F	T	F	F	F	T	F
i7	F	F	T	T	T	T	T	T	F
i8	F	F	F	T	F	F	F	T	F

- **i1,i2,i5,i7** are *models* of U and **i3,i4,i6,i8** are *anti-models* of U
- $W(p,q,r)$ is a *tautology*, all its 8 interpretations are also its models
- $Z(p,q,r)$ is *inconsistent*, it is evaluated as false in all its 8 interpretations
- $U \equiv V$, U and V are *logically equivalent* because they have identical truth tables.
- $U \models \neg p \vee q$ (*logical consequence*) because in all interpretations (**i1,i2,i5,i7**) which evaluate the formula U as true, the formula $\neg p \vee q$ is also evaluated as true.

Logical equivalences

Simplification laws:

$$\begin{array}{ll} \neg\neg U \equiv U & \text{and } U \rightarrow U \equiv T \\ U \wedge \neg U \equiv F & \text{and } U \vee \neg U \equiv T \\ T \wedge U \equiv U & \text{and } F \vee U \equiv U \\ U \rightarrow T \equiv T & \text{and } U \rightarrow F \equiv \neg U \\ T \rightarrow U \equiv U & \text{and } F \rightarrow U \equiv T \\ U \leftrightarrow T \equiv U & \text{and } U \leftrightarrow F \equiv \neg U \\ U \oplus T \equiv \neg U & \text{and } U \oplus F \equiv U \\ U \leftrightarrow U \equiv T & \text{and } U \oplus U \equiv F \end{array}$$

Commutative laws:

$$\begin{array}{l} U \wedge V \equiv V \wedge U \\ U \vee V \equiv V \vee U \end{array}$$

Distributive laws:

$$\begin{array}{l} U \wedge (V \vee Z) \equiv (U \wedge V) \vee (U \wedge Z) \\ U \vee (V \wedge Z) \equiv (U \vee V) \wedge (U \vee Z) \end{array}$$

Logical equivalences (contd.)

Idempotency laws:

$$U \wedge U \equiv U$$

$$U \vee U \equiv U$$

Absorption laws:

$$U \wedge (U \vee V) \equiv U$$

$$U \vee (U \wedge V) \equiv U$$

De Morgan laws:

$$\neg(U \wedge V) \equiv \neg U \vee \neg V$$

$$\neg(U \vee V) \equiv \neg U \wedge \neg V$$

Associative laws:

$$(U \wedge V) \wedge Z \equiv U \wedge (V \wedge Z)$$

$$(U \vee V) \vee Z \equiv U \vee (V \vee Z)$$

Logical equivalences (contd.)

--- Definitions of the connectives ---

$$U \rightarrow V \equiv \neg U \vee V$$

$$U \rightarrow V \equiv U \leftrightarrow (U \wedge V)$$

$$U \leftrightarrow V \equiv (U \rightarrow V) \wedge (V \rightarrow U)$$

$$U \leftrightarrow V \equiv (U \vee V) \rightarrow (U \wedge V)$$

$$U \vee V \equiv \neg(\neg U \wedge \neg V)$$

$$U \vee V \equiv \neg U \rightarrow V$$

$$\neg U \equiv U \uparrow U$$

$$U \vee V \equiv (U \uparrow U) \uparrow (V \uparrow V)$$

$$U \vee V \equiv (U \downarrow V) \downarrow (U \downarrow V)$$

$$U \rightarrow V \equiv \neg(U \wedge \neg V)$$

$$U \rightarrow V \equiv V \leftrightarrow (U \vee V)$$

$$U \oplus V \equiv \neg(U \rightarrow V) \vee \neg(V \rightarrow U)$$

$$U \wedge V \equiv \neg(\neg U \vee \neg V)$$

$$U \wedge V \equiv \neg(U \rightarrow \neg V)$$

$$\neg U \equiv U \downarrow U$$

$$U \wedge V \equiv (U \downarrow U) \downarrow (V \downarrow V)$$

$$U \wedge V \equiv (U \uparrow V) \uparrow (U \uparrow V)$$

➤ A set of connectives is **functionally complete** if there is no truth table that can not be expressed as a formula involving only these connectives. All the other connectives can be expressed using the connectives from the set.

➤ The following sets of connectives are **functionally complete**.

- | | | | |
|------------------------|--------------------------|-----------------------------|-------------------------------|
| 1. $\{\neg, \wedge\};$ | 2. $\{\neg, \vee\};$ | 3. $\{\neg, \rightarrow\};$ | 4. $\{\uparrow\};$ |
| 5. $\{\downarrow\};$ | 6. $\{\oplus, \wedge\};$ | 7. $\{\oplus, \vee\};$ | 8. $\{\oplus, \rightarrow\};$ |

➤ **The duality principle:**

For every logical equivalence $U \equiv V$ containing only the connectives: $\neg, \wedge, \vee, \uparrow, \downarrow, \leftrightarrow, \otimes$ there is another logical equivalence $U' \equiv V'$, where U', V' are formulas obtained from U, V by interchanging the connectives $(\wedge, \vee), (\uparrow, \downarrow), (\leftrightarrow, \otimes)$ and the truth values (T, F) .

➤ Notice that some of the above laws are pairs of **dual logical equivalences**!

➤ **Dual connectives:** $(\wedge, \vee), (\uparrow, \downarrow), (\leftrightarrow, \otimes)$.

➤ **Dual truth values:** (T, F) .

Sets of propositional formulas

Definition:

- The set $\{U_1, U_2, \dots, U_n\}$ is called **consistent** if

the formula $U_1 \wedge U_2 \wedge \dots \wedge U_n$ is consistent:

$$\exists i : F_P \rightarrow \{T, F\} \text{ such that } i(U_1 \wedge U_2 \wedge \dots \wedge U_n) = T.$$

- The set $\{U_1, U_2, \dots, U_n\}$ is called **inconsistent** if

the formula $U_1 \wedge U_2 \wedge \dots \wedge U_n$ is inconsistent:

$$\forall i : F_P \rightarrow \{T, F\}, i(U_1 \wedge U_2 \wedge \dots \wedge U_n) = F.$$

- The formula V is a **logical consequence** of the set $\{U_1, U_2, \dots, U_n\}$ of formulas,

notation: $U_1, U_2, \dots, U_n \models V$, if

$$\forall i : F_P \rightarrow \{T, F\} \text{ such that } i(U_1 \wedge U_2 \wedge \dots \wedge U_n) = T, \text{ we have } i(V) = T.$$

The formulas $U_1, U_2, \dots, U_n$ are called **premises, hypotheses, facts**, and

V is called **conclusion**.

Theorems (semantic results)

Theorem 1: Let $U_1, U_2, \dots, U_n, U, V$ be propositional formulas.

- $\models U$ if and only if $\neg U$ is inconsistent.
- $U \models V$ if and only if $\models U \rightarrow V$ if and only if $\{U, \neg V\}$ is inconsistent.
- $U \equiv V$ if and only if $\models U \leftrightarrow V$.
- $U_1, U_2, \dots, U_n \models V$ if and only if $\models U_1 \wedge U_2 \wedge \dots \wedge U_n \rightarrow V$ if and only if the set $\{U_1, U_2, \dots, U_n, \neg V\}$ is inconsistent.

Theorem 2: Let $S = \{U_1, U_2, \dots, U_n\}$ be a set of propositional formulas.

1. If S is a consistent set, then $\forall j, 1 \leq j \leq n, S - \{U_j\}$ is a consistent set.
2. If S is a consistent set and V is a valid formula, then $S \cup \{V\}$ is consistent.
3. If S is an inconsistent set, then $\forall V \in F_P, S \cup \{V\}$ is inconsistent.
4. If S is an inconsistent set and U_j is valid, where $1 \leq j \leq n$, then $S - \{U_j\}$ is inconsistent.

Example

A client describes the requirements of a software application:

- R_1 . *If condition A is satisfied then condition B must also be satisfied.*
- R_2 . *If conditions B and C are satisfied, then D must also be satisfied.*
- R_3 . *If condition D is satisfied then condition A is not satisfied.*
- R_4 . *If condition C is satisfied then A must also be satisfied.*
- R_5 . *If A is satisfied then D or C are satisfied.*
- R_6 . *C is satisfied if neither B nor A are satisfied.*
- R_7 . *B is not satisfied if C is not satisfied.*

Are these requirements simultaneously satisfiable?

In order to answer the question we have to check the consistency/inconsistency of the formula:

$$U = R_1 \wedge R_2 \wedge R_3 \wedge R_4 \wedge R_5 \wedge R_6 \wedge R_7.$$

Example (contd.)

- R_1 . If condition A is satisfied *then* condition B must also be satisfied.
 R_2 . If conditions B and C are satisfied, *then* D must also be satisfied.
 R_3 . If condition D is satisfied *then* condition A is *not* satisfied.
 R_4 . If condition C is satisfied *then* A must also be satisfied.
 R_5 . If A is satisfied *then* D or C are satisfied.
 R_6 . C is satisfied *if neither* B nor A are satisfied.
 R_7 . B is *not* satisfied *if* C is *not* satisfied.

$$R_1. A \rightarrow B \equiv \neg A \vee B$$

$$R_2. B \wedge C \rightarrow D \equiv \neg B \vee \neg C \vee D$$

$$R_3. D \rightarrow \neg A \equiv \neg D \vee \neg A$$

$$R_4. C \rightarrow A \equiv \neg C \vee A$$

$$R_5. A \rightarrow C \vee D \equiv \neg A \vee C \vee D$$

$$R_6. \neg A \wedge \neg B \rightarrow C \equiv A \vee B \vee C$$

$$R_7. \neg C \rightarrow \neg B \equiv C \vee \neg B$$

$$U = R_1 \wedge R_2 \wedge R_3 \wedge R_4 \wedge R_5 \wedge R_6 \wedge R_7.$$

Example (contd.) – Truth table

	A	B	C	D	R_1	R_2	R_3	R_4	R_5	R_6	R_7	U
i_1	T	T	T	T	T	T	F	T	T	T	T	F
i_2	T	T	T	F	T	F	T	T	T	T	T	F
i_3	T	T	F	T	T	T	F	T	T	T	F	F
i_4	T	T	F	F	T	T	T	T	F	T	F	F
i_5	T	F	T	T	F	T	F	T	T	T	T	F
i_6	T	F	T	F	F	T	T	T	T	T	T	F
i_7	T	F	F	T	F	T	F	T	T	T	T	F
i_8	T	F	F	F	F	T	T	T	F	T	T	F
i_9	F	T	T	T	T	T	T	F	T	T	T	F
i_{10}	F	T	T	F	T	F	T	F	T	T	T	F
i_{11}	F	T	F	T	T	T	T	T	T	T	F	F
i_{12}	F	T	F	F	T	T	T	T	T	T	F	F
i_{13}	F	F	T	T	T	T	T	F	T	T	T	F
i_{14}	F	F	T	F	T	T	T	F	T	T	T	F
i_{15}	F	F	F	T	T	T	T	T	T	F	T	F
i_{16}	F	F	F	F	T	T	T	T	T	F	T	F

The column of U contains only the truth value F , so U is evaluated as false in all 16 interpretations, therefore U is an inconsistent formula.

The conclusion is that *the requirements are contradictory, they cannot be satisfied simultaneously by the software application.*

Normal forms - *definitions*

A literal is a propositional variable or its negation.	$p, \neg q, r$
A clause is a disjunction of a finite number of literals.	$p, \neg p \vee q, r \vee q \vee s$
A cube is a conjunction of a finite number of literals.	$q, p \wedge \neg q, r \wedge s \wedge p$
A formula is in disjunctive normal form (DNF), if it is written as a disjunction of cubes: $\bigvee_{i=1}^p \left(\bigwedge_{j=1}^q l_{ij} \right)$	$p \vee \neg q \vee r$ - 3 unit cubes $p \wedge q$ - DNF, 1 cube $p \vee (q \wedge r) \vee (\neg p \wedge \neg r \wedge s)$ - 3 cubes
A formula is in conjunctive normal form (CNF), if it is written as a conjunction of clauses: $\bigwedge_{i=1}^n \left(\bigvee_{j=1}^m l_{ij} \right)$	$p \vee \neg q \vee r$ - CNF, 1 clause $p \wedge q$ - 2 unit clauses $p \wedge (q \vee r) \wedge (\neg p \vee \neg r \vee s)$ - 3 clauses

Property

Let $\{l_1, l_2, \dots, l_n\}$ be a set of literals.

The following sentences are equivalent:

- a) The clause $\bigvee_{i=1}^n l_i$ is a tautology;
- b) The cube $\bigwedge_{i=1}^n l_i$ is inconsistent;
- c) The set $\{l_1, l_2, \dots, l_n\}$ of literals contains at least one pair of opposite literals:
 $\exists i, j \in \{1, \dots, n\}$ such that $l_j = \neg l_i$.

Examples

- the **clause** $U = \underline{p} \vee q \vee r \vee \underline{\neg p}$ is a **tautology** ($U \equiv T$), because $p, \neg p$ are opposite literals.
- the **cube** $V = \underline{p} \wedge q \wedge r \wedge \underline{\neg p}$ is an **inconsistent** formula ($V \equiv F$), because $p, \neg p$ are opposite literals.

Normalization algorithm

Aim: to transform a formula into another logically equivalent formula, having a certain character of “normal” or “canonical” form.

Transformations which **preserve the logical equivalence** are applied:

Step1: The formulas of “ $X \rightarrow Y$ ” type are replaced by the equivalent form $\neg X \vee Y$.

The formulas of “ $X \leftrightarrow Y$ ” type are replaced by the equivalent form

$$(\neg X \vee Y) \wedge (\neg Y \vee X).$$

Step2: De Morgan laws are applied $\implies$ push negations in until they apply only to propositional variables

Multiple negations are eliminated by the reduction rule: $\neg \neg X \equiv X$.

Step3: The distribution laws are applied.

Normal forms – *theoretical results*

Theorem 3:

Every propositional formula admits an equivalent CNF and an equivalent DNF.

Theorem 4:

1. A formula in CNF is a **tautology** if and only if **all its clauses are tautologies**.
2. A formula in DNF is **inconsistent** if and only if **all its cubes are inconsistent**.

Remarks:

- The first part of the above theorem provides a **direct method** to prove that a formula is a tautology.
- DNF of a propositional formula **provides all the models of that formula**, finding all the interpretations that evaluate the cubes as true.
- CNF of a propositional formula **provides all the anti-models of that formula**, finding all the interpretations that evaluate, one by one, the clauses as false.

Dual concepts: clause-cube, DNF-CNF.

Example

Write the equivalent DNF of the formula and find its models.

$$X = (p \wedge q \rightarrow r) \rightarrow (p \rightarrow r) \wedge q$$

We apply the normalization algorithm:

$$X = (p \wedge q \overset{1}{\rightarrow} r) \overset{2}{\rightarrow} (p \overset{3}{\rightarrow} r) \wedge q \quad (\text{replace } \overset{2}{\rightarrow} \text{ using } U \rightarrow V \equiv \neg U \vee V)$$

$$\equiv \neg(p \wedge q \overset{1}{\rightarrow} r) \vee (p \overset{3}{\rightarrow} r) \wedge q \quad (\text{replace } \overset{1,3}{\rightarrow} \text{ using } U \rightarrow V \equiv \neg U \vee V)$$

$$\equiv \neg(\neg(p \wedge q) \vee r) \vee (\neg p \vee r) \wedge q \quad (\text{apply de Morgan's law})$$

$$\equiv (p \wedge q \wedge \neg r) \vee (\neg p \vee r) \wedge q \quad (\text{apply distribution of } \wedge \text{ over } \vee)$$

$$\equiv (p \wedge q \wedge \neg r) \vee (\neg p \wedge q) \vee (r \wedge q) \quad - \text{DNF with 3 cubes}$$

The **models** of X are the interpretations that evaluate one by one the **cubes** of DNF as true.

Example – *models of a formula*

$$\text{DNF}(X) = (p \wedge q \wedge \neg r) \vee (\neg p \wedge q) \vee (r \wedge q)$$

Cube: $p \wedge q \wedge \neg r$ provides one model:

$$i1: \{p, q, r\} \rightarrow \{T, F\}, i1(p)=T, i1(q)=T, i1(r)=F$$

Cube: $\neg p \wedge q$ provides 2 models:

$$i2: \{p, q, r\} \rightarrow \{T, F\}, i2(p)=F, i2(q)=T, i2(r)=T$$

$$i3: \{p, q, r\} \rightarrow \{T, F\}, i3(p)=F, i3(q)=T, i3(r)=F$$

Cube: $r \wedge q$ provides 2 models:

$$i4: \{p, q, r\} \rightarrow \{T, F\}, i4(p)=T, i4(q)=T, i4(r)=T$$

$$i5: \{p, q, r\} \rightarrow \{T, F\}, i5(p)=F, i5(q)=T, i5(r)=T$$

We notice that $i2 = i5$.

The **models** of X are the interpretations: $i1, i2, i3, i4$.

$$i1(X) = i2(X) = i3(X) = i4(X) = T$$

All the other four interpretations evaluate the formula X as false, they are **anti-models** of X.